realized skewness, volatility ratio between time resolutions, direction change indicators, and overlapping returns. A word of caution. A cursory glance at financial markets will display the complexity of simply inquiring about the price. Price quotes could be the bid-ask pair; transaction prices, which may or may not be former bid or ask quotes; irregular bid, ask, transaction prices; or simply the mid-price. The research process needs to be clear in its acquisition of the correct data needed, as trivial requests may lead to adverse results.

Having acquired high frequency data we are now ready for the analysis. The preferred methodology for research is the scientific method which consists of three steps (Dacorogna et al., 2001). The first of these steps is to discover and understand the fundamental, statistical properties of the data. These properties are known as "stylized facts" in the econometrics and finance literature and are quite widely reported. See, for example, Cont (2001) for an empirical analysis of asset returns and their associated stylized facts and statistical properties.

Secondly, we formulate models based on empirical facts. The focus here should be on robust models derived from empirical analysis rather than on loose conjecture. This is where our understanding of the markets and the data should coincide. There has been quite some debate on whether to take a time-series (modeling statistical properties of the data) or microstructure (modeling market behavior) approach, though with high frequency data one should be able to test microstructure models (Hasbrouck, 2007; Dacorogna et al., 2001). The third and final step is to test whether these models reproduce the stylized facts and statistical properties discovered in the first step. The focus here is on prediction of future movements, opportunities, risks and rewards. It is thus clear that to undertake this process successfully, we need good, reliable high frequency data.

STRATEGIES

Statistical Arbitrage

If you can simultaneously buy an asset low and trade it high you have pure, deterministic arbitrage. Existence of deterministic arbitrage lies around the idea that a riskless profit can be made by replicating the future payoffs of an asset with a basket of other assets, with the price of the replicating portfolio not exceeding the costs of the original asset. This can be formalized thus:

$$|\text{payoff}(S_t - R(S_t))| > \text{TransactionCost} \tag{13.1}$$